

LA-4

classmate

Date _____

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Loss minimizatⁿ in classificatⁿ

Recap

why vectors?

Answer - to study higher dimension

hyperplane eqⁿ

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_N x_N + w_0 = 0$$

can be represented in vector $w^T X + w_0 = 0$

Recap

(a) vectors: $\bar{x} \in \mathbb{R}^d$ $\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$

(b) norm: $\|\bar{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2}$

(c) Inner product (dot product) $(\bar{x} \cdot \bar{y})$

$$x, y \in \mathbb{R}^d$$

$$(\bar{x} \cdot \bar{y}) = \bar{x}^T \bar{y} = \sum_{i=1}^d x_i y_i$$

(d) angle betⁿ vector

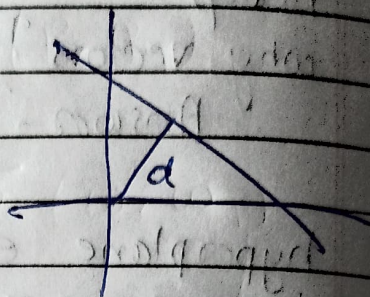
$$\bar{x}^T \bar{y} = \|\bar{x}\| \|\bar{y}\| \cos \theta$$

$$\cos \theta = \frac{\bar{x}^T \bar{y}}{\|\bar{x}\| \|\bar{y}\|}$$

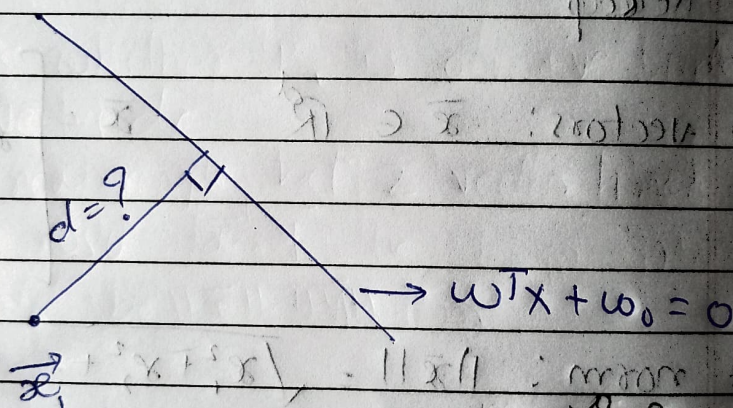
(c) distance betⁿ origin & line

$$\omega^T \bar{x} + \omega_0 = 0$$

$$d = \left| \frac{\omega_0}{\|\omega\|} \right|$$



Distance of HP From any point.



$$d = \left| \frac{\omega^T x_i + \omega_0}{\|\omega\|} \right|$$

Need to
study this
formula
this
derived

same

if $\bar{x}_i$ is $(0,0) \rightarrow$ origin then

$$d = \left| \frac{\omega^T \cdot 0 + \omega_0}{\|\omega\|} \right|$$

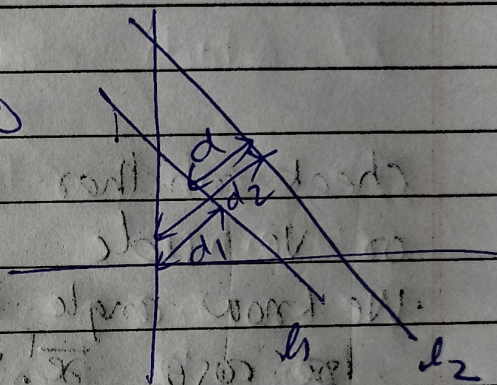
Distance betⁿ 2 planes

if 2 planes are not parallel then they must be intersecting somewhere. So, distance is 0

if 2 planes are parallel then distance will be > 0

$$L_1 = \omega^T x + \beta = 0$$

$$L_2 = \omega^T x + \bar{\beta} = 0$$



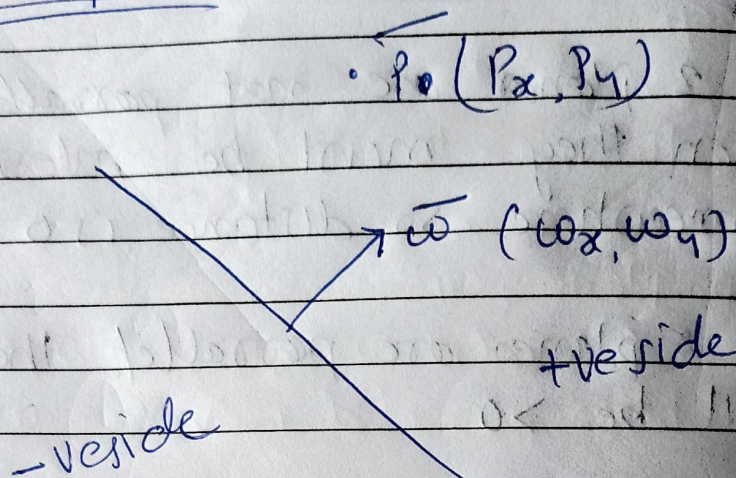
$$d = d_1 - d_2$$

$$= \frac{3}{\|w\|} - \frac{5}{\|w\|}$$

$$= \frac{2}{\|w\|}$$

$$= \frac{\omega_0'^2 - \omega_0^2}{\|w\|}$$

Halfspaces



$$w^T x + w_0 \geq 0$$

check whether p_i is on +ve side or -ve side

We know angle betⁿ 2 ~~vector~~ lines

$$\cos \theta = \frac{\bar{x}^T \bar{y}}{\|\bar{x}\| \|\bar{y}\|}$$

similarly angle betⁿ $\bar{p}$ & $\bar{w}$

$$\cos \theta = \frac{\bar{w}^T \bar{p}}{\|\bar{w}\| \|\bar{p}\|}$$

we know from figure above for $\odot$ +ve side $\odot$ θ values will be ~~either~~ either $0-90^\circ$ or $270^\circ-360^\circ$

for -ve side θ value will be ~~either~~ $90^\circ-270^\circ$

$270^\circ - 360^\circ$ will be +ve (>0)

& ~~the~~ cost for $90^\circ - 270^\circ$ will be -ve (<0)

~~hence~~ $\|\vec{w}\|$ & $\|\vec{p}\|$ will be always +ve (1)

hence we can say that

$$\vec{w}^T \cdot \vec{p} = \|\vec{w}\| \|\vec{p}\| \cos \theta$$

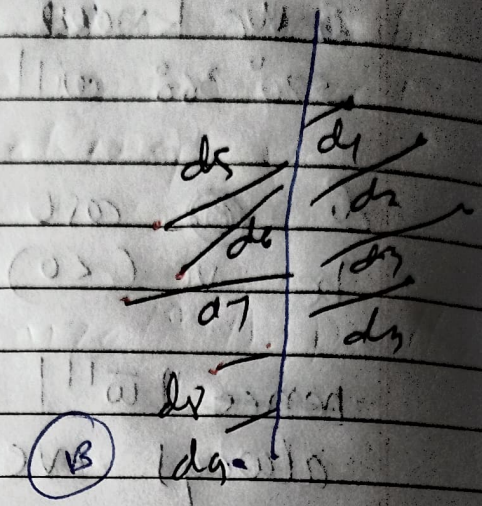
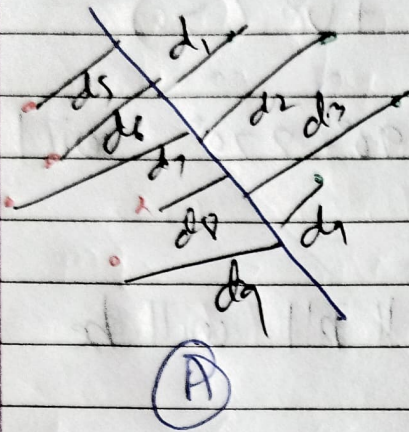
$\vec{w}^T \cdot \vec{p} > 0$ when θ is 0° or $270^\circ - 360^\circ$

$\vec{w}^T \cdot \vec{p} < 0$ when θ is $90^\circ - 270^\circ$

→ This is +ve side

→ This is -ve side

loss function



Which classifier is better

If we want to find best classifier

the separation or distance of points from the hyperplane should be maximum.

Gain function

$$= d_1 + d_2 + \dots$$

best classifier where gain function is maximum

Suppose $X = \{x_i; y_i\}_{i=1}^n$

$$\begin{array}{c|c} x_1 & y \\ \hline x_2 & +1 \\ & -1 \end{array}$$

$w^T x + w_0 = 0$ - general linear

Gain function

$$G(x, \vec{w}, w_0) = \sum_{i=1}^n y_i \left(\frac{w^T x_i + w_0}{\|w\|} \right)$$

best classifier

$$\max_{\vec{w}, w_0} G(x, \vec{w}, w_0)$$

$$\max_{\vec{w}, w_0} \left(\sum_{i=1}^n \left(\frac{w^T \bar{x}_i + w_0}{\|w\|} \right) y_i \right)$$

loss function = - Gain function

maximization problem can be converted into minimization problem by taking -ve.

ML $\rightarrow$ minimization is preferred

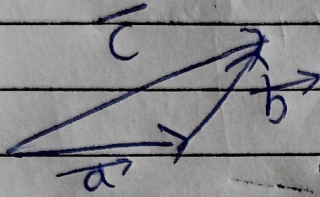
Loss function = 0

$$\min_{\bar{w}, w_b} \left(- \sum \left(\frac{\bar{w}^T x_i + w_b}{\|w\|} \right) y_i \right)$$

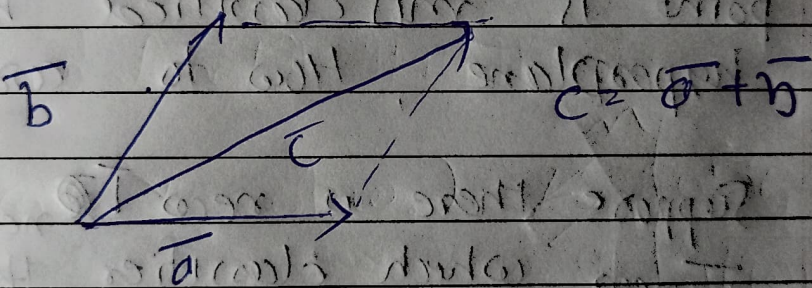
Some Vector laws

① Triangle law of add

$$\vec{c} = \vec{a} + \vec{b}$$

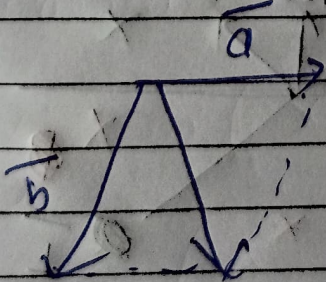


② parallelogram law



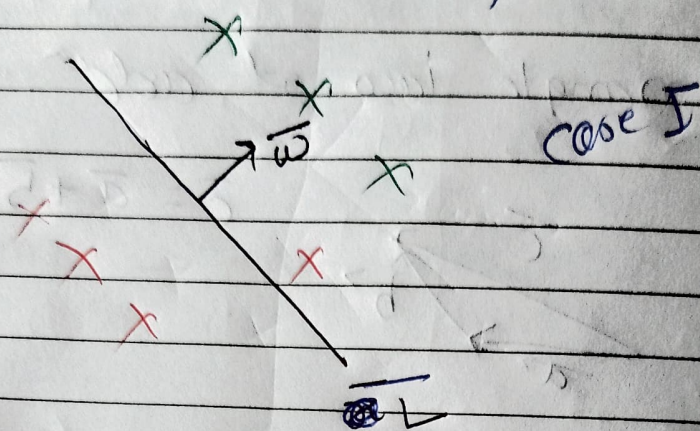
$$\vec{c} = \vec{a} + \vec{b}$$

$$\vec{c} = \vec{a} - \vec{b}$$



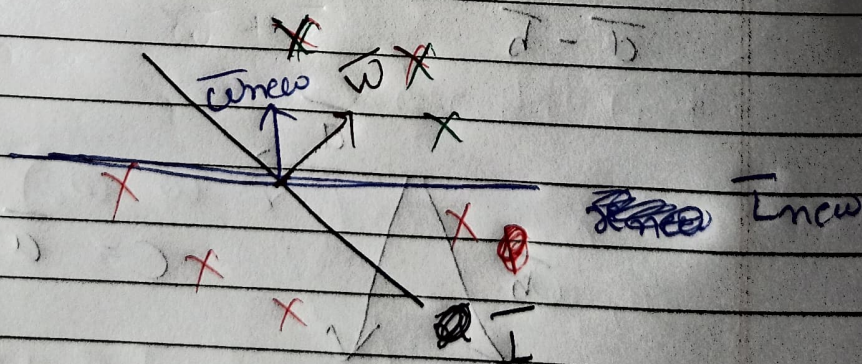
$$\vec{c} = \vec{a} - \vec{b}$$

Perceptron learning algorithm

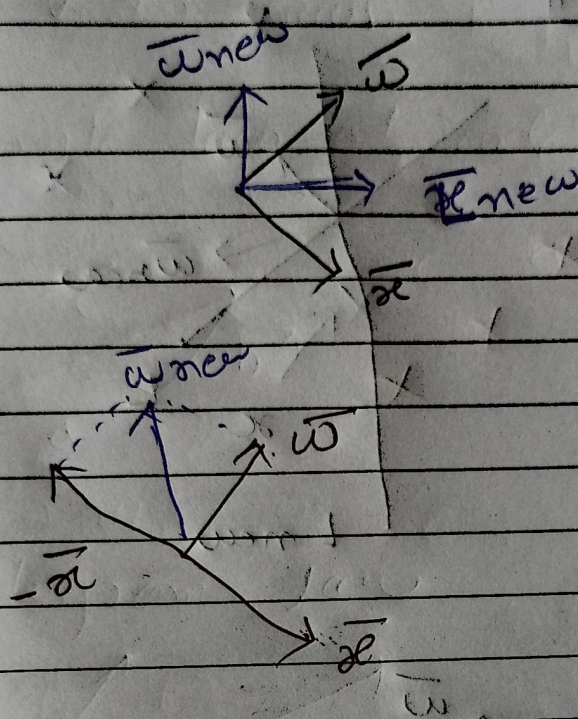


As shown in figure the red point is misclassified by $\vec{L}$ hyperplane. How to correct it?

Suppose there is new $\vec{L}_{new}$ & $\vec{w}_{new}$ which classifies the all points correctly



Let's call misclassified red point x

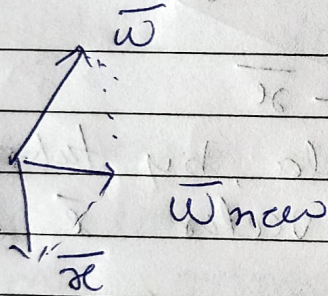
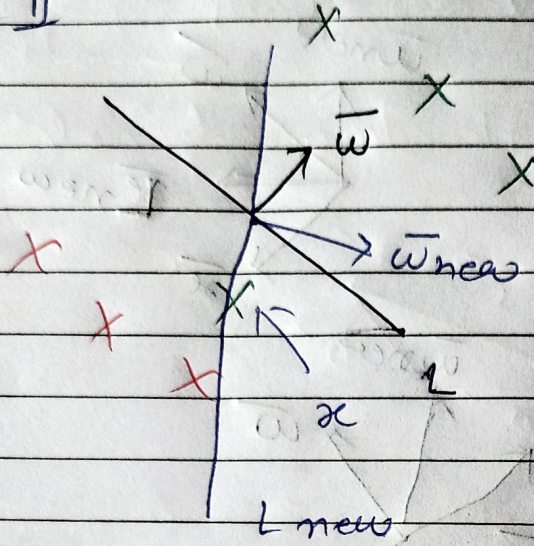


$$\vec{w}_{new} = \vec{w} - \vec{x}$$

above formula, by taking mis
classified ~~vector~~ vector $\vec{x}$ & weight
vector $\vec{w}$

$$\vec{x} + \vec{w}$$

case II



$$\bar{w}_{new} = \bar{w} + \bar{x}$$

Summary

$$(1) \bar{w}_{new} = \bar{w} + \bar{x}$$

x is misclassified | clockwise

Actual $\rightarrow +1$

assigned $\rightarrow -1$

$$(2) \bar{w}_{new} = \overline{\bar{w} + \bar{x}}$$

$$\bar{w} - \bar{x}$$

x is misclassified | Anti clockwise

Actual $\rightarrow -1$

assigned $\rightarrow +1$

$$w_{new} = w + y \vec{x}$$

perceptron update rule.